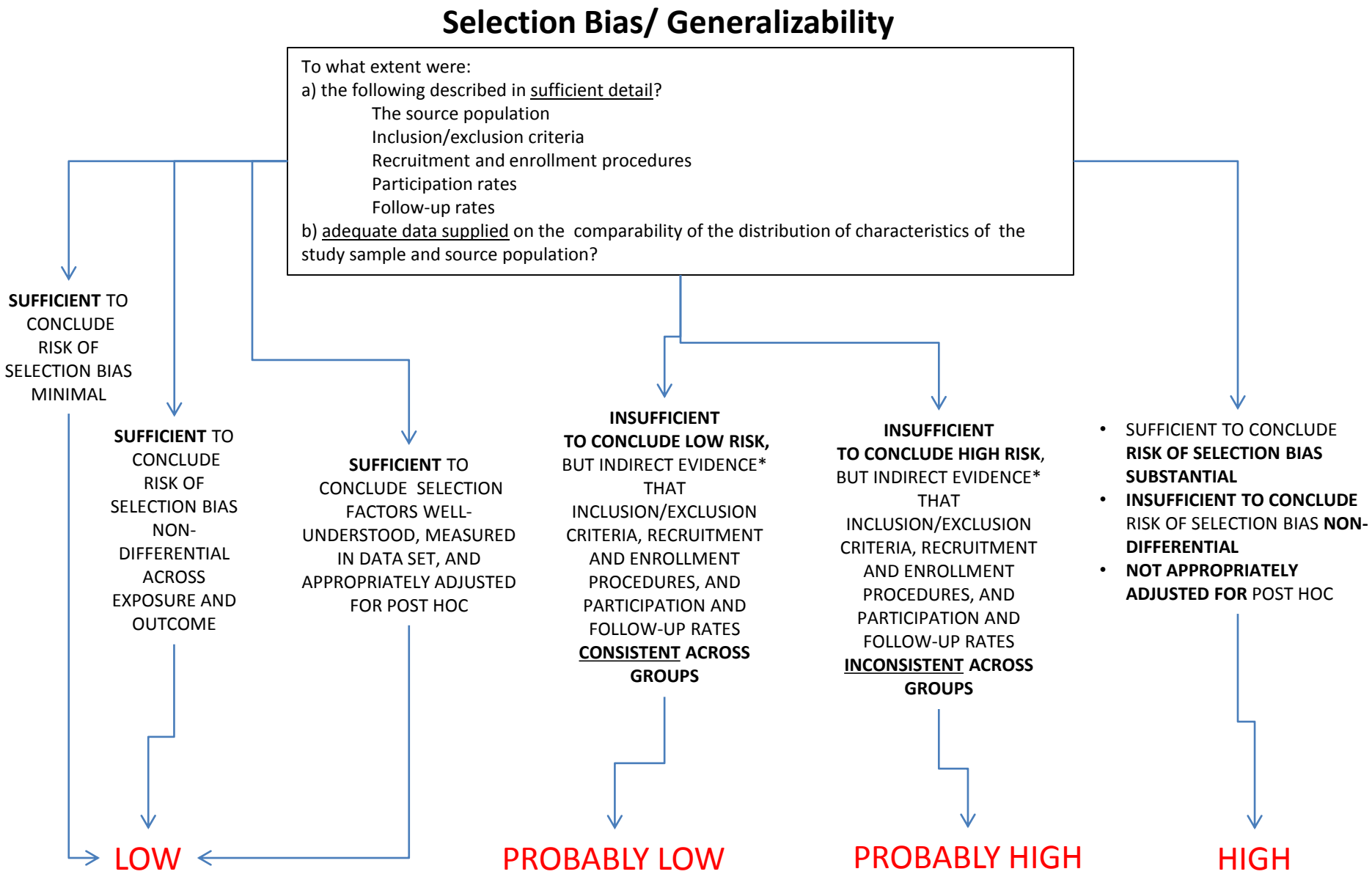
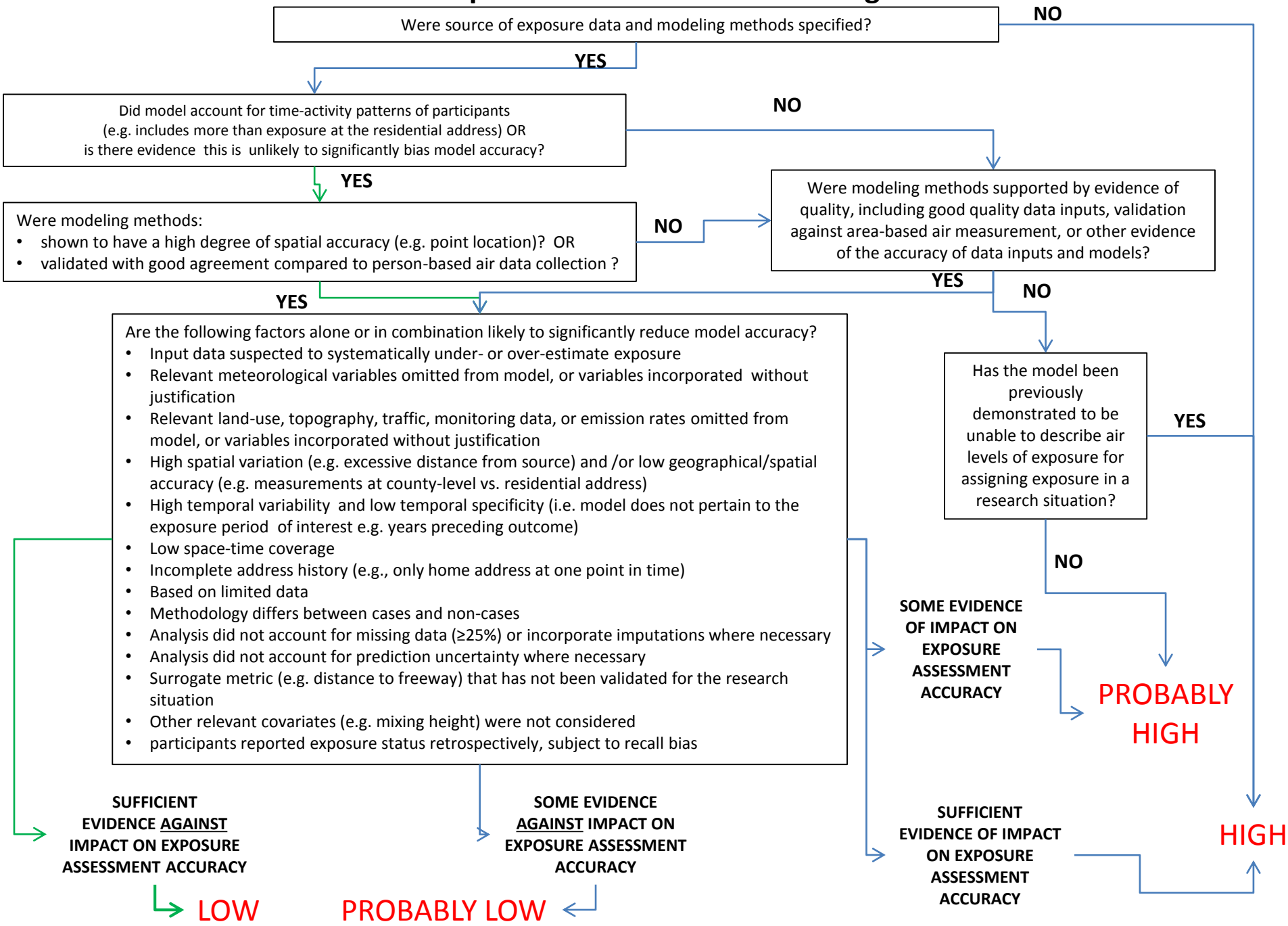
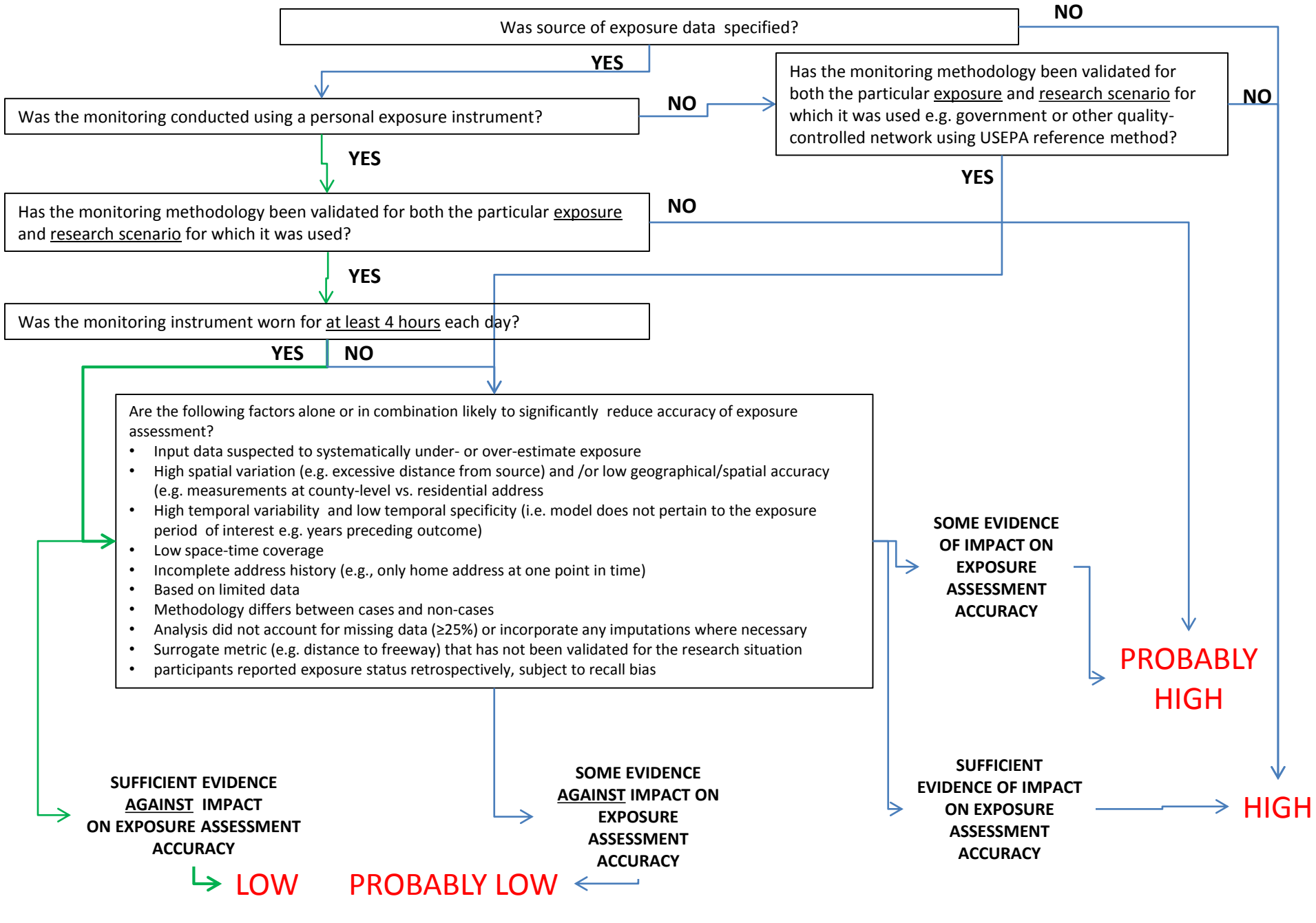
Online supplementary figure S1. Risk of bias criteria adapted from Lam J, Sutton P, Kalkbrenner A, et al. A Systematic Review and Meta-Analysis of Multiple Airborne Pollutants and Autism Spectrum Disorder. PLoS One. 2016;11(9):e0161851. Published 2016 Sep 21. doi:10.1371/journal.pone.0161851



Exposure Assessment - Modeling



Exposure Assessment - Monitoring



Confounding

To what extent did the study measure <u>important</u> potential confounders consistently across study groups using valid and reliable methods and account for them in design or analysis:

- **<u>Age</u>**
- **<u>Sex</u>**
- **<u>Smoking</u>**
- Alcohol consumption
- Race
- Diet
- Medication
- Comorbidity
- Obesity (BMI)
- Education
- Occupation
- Income
- **<u>Individual or Neighborhood SES</u>**
- Spatial autocorrelation
- Greenness, noise

<u>OR</u> did the study report that potential confounders were <u>evaluated and omitted</u> because inclusion did not substantially affect the results?

ALL IMPORTANT CONFOUNDERS
e.g. includes all of age, sex, smoking, SES

→ LOW

MOST IMPORTANT CONFOUNDERS AND THIS IS NOT EXPECTED TO INTRODUCE SUBSTANTIAL BIAS
e.g. indirect adjustment for omitted covariates

→ PROBABLY LOW

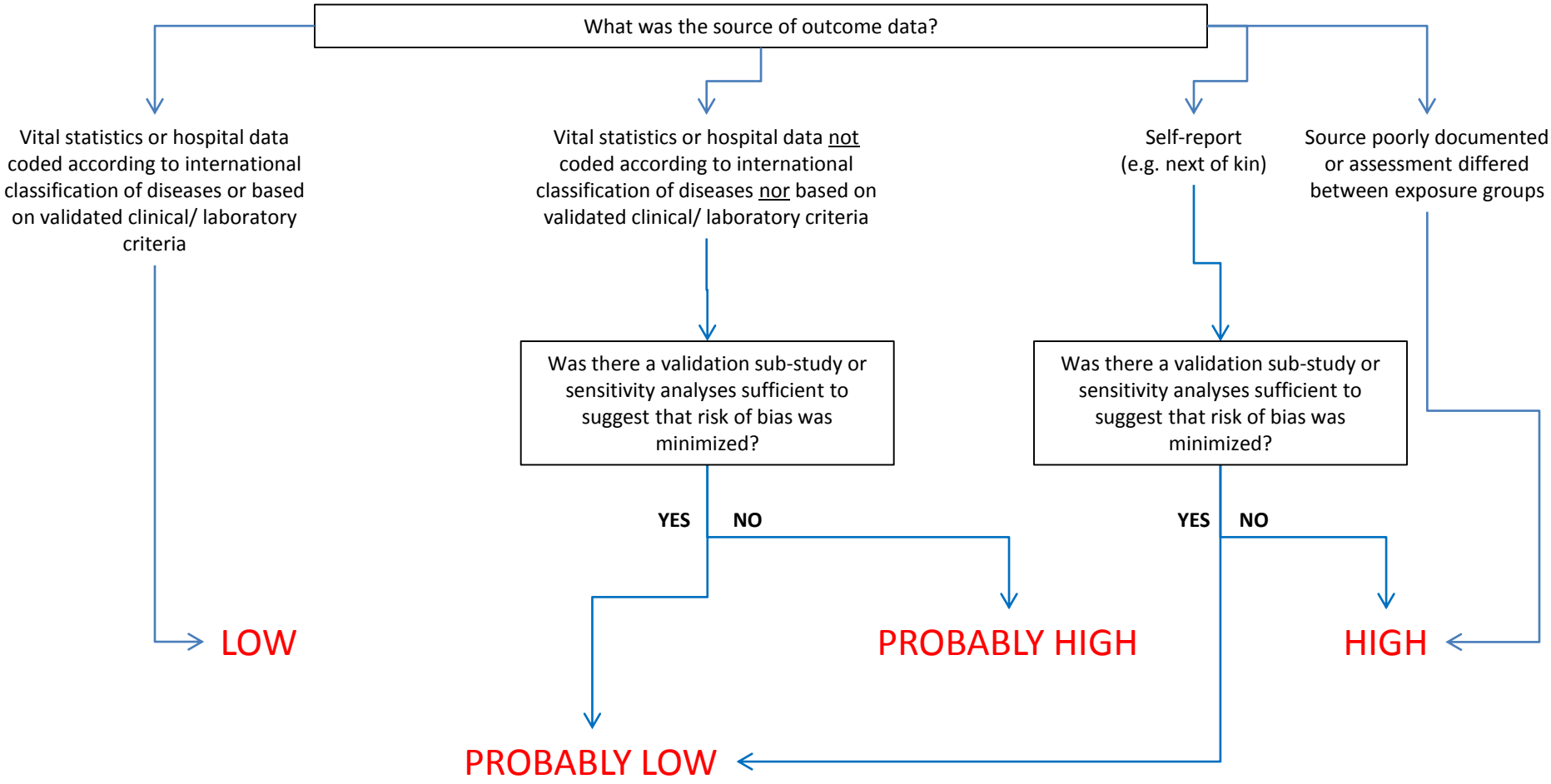
SOME BUT NOT ALL IMPORTANT CONFOUNDERS AND THIS IS EXPECTED TO INTRODUCE SUBSTANTIAL BIAS
e.g. missing any 1 of age, sex, smoking, SES

→ PROBABLY HIGH

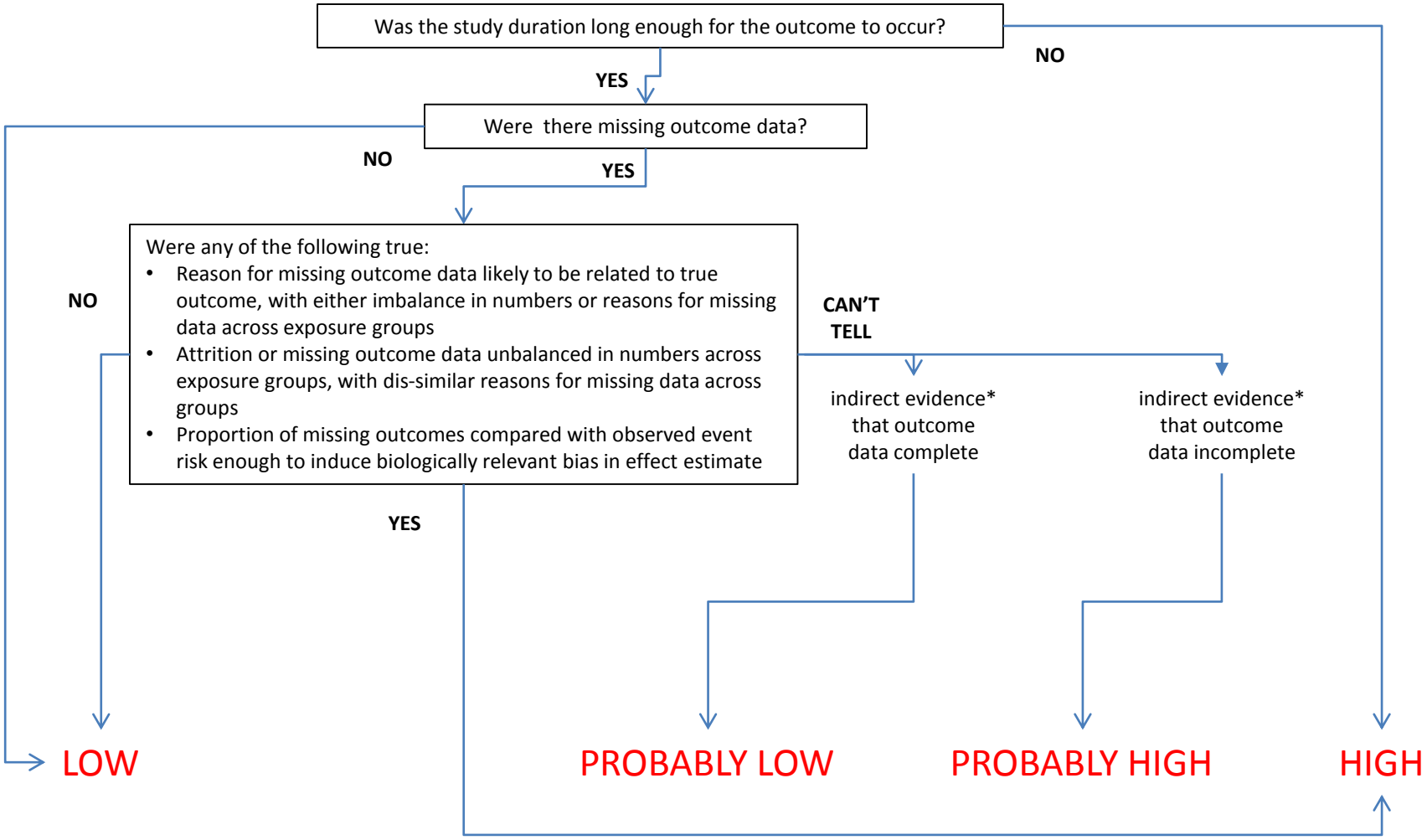
DID NOT ACCOUNT FOR OR EVALUATE MULTIPLE IMPORTANT CONFOUNDERS
e.g. missing 2 or more of age, sex, smoking, SES

→ HIGH

Outcome Assessment

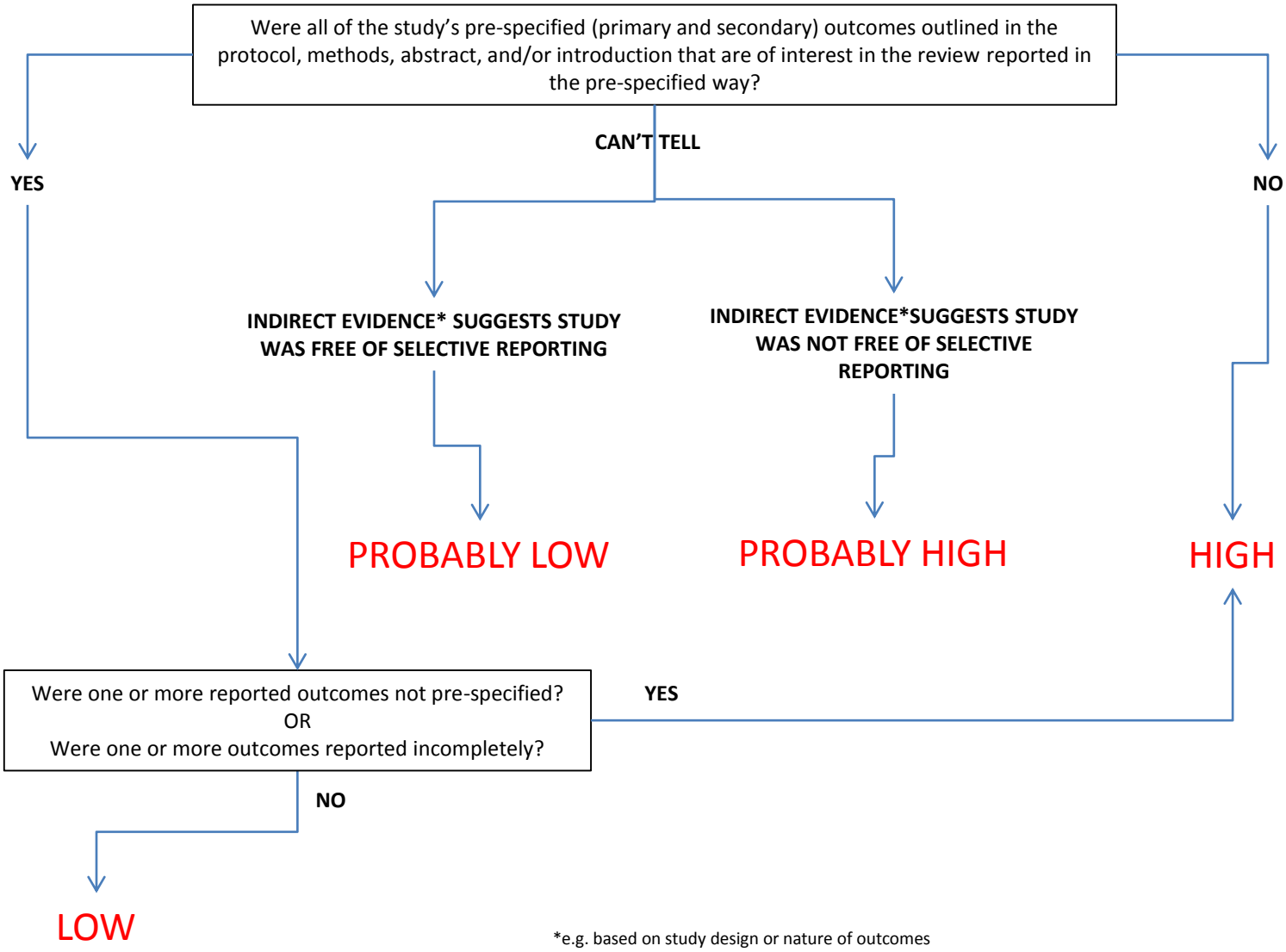


Completeness of Outcome Data



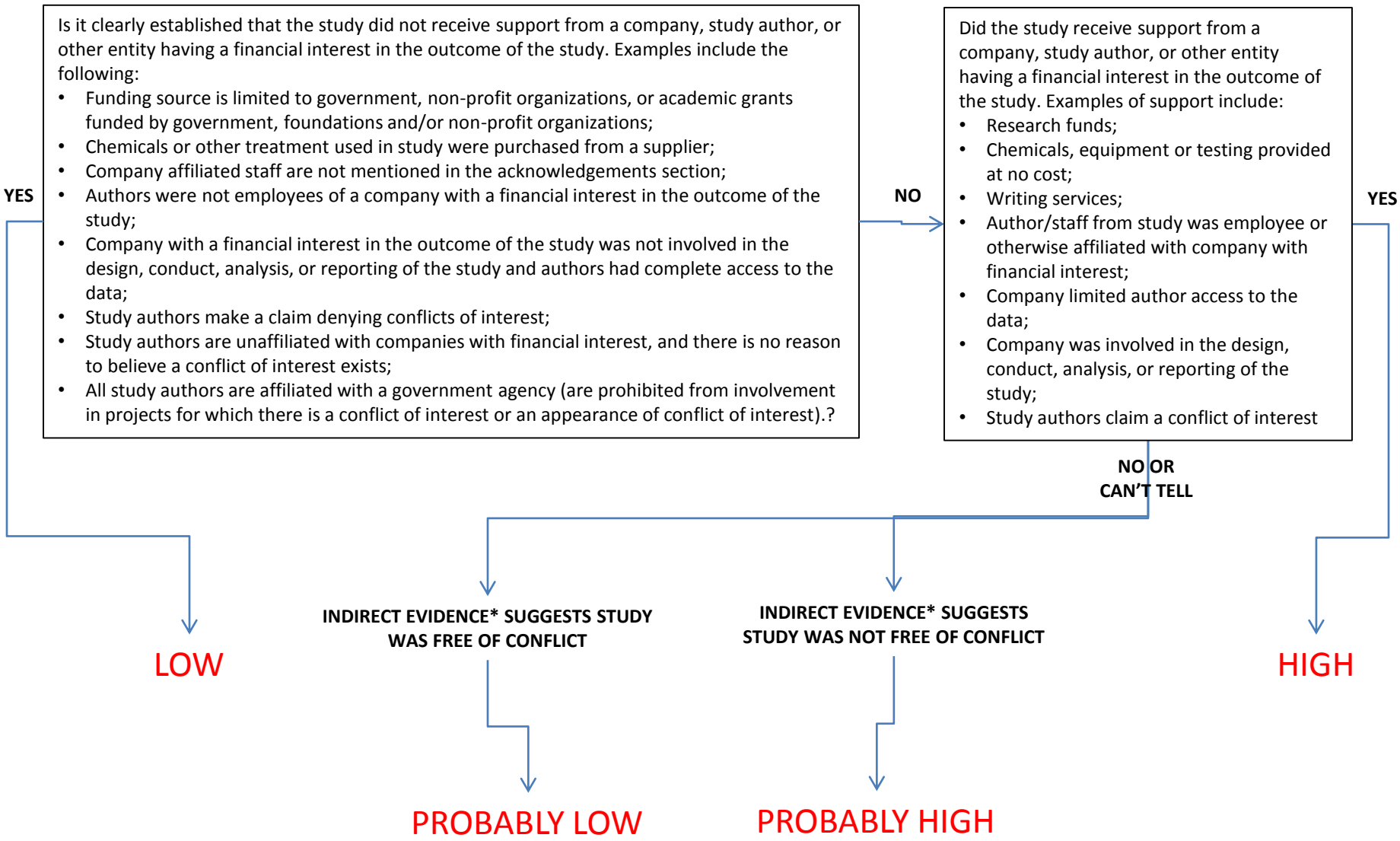
*e.g. based on study design

Selective Outcome Reporting



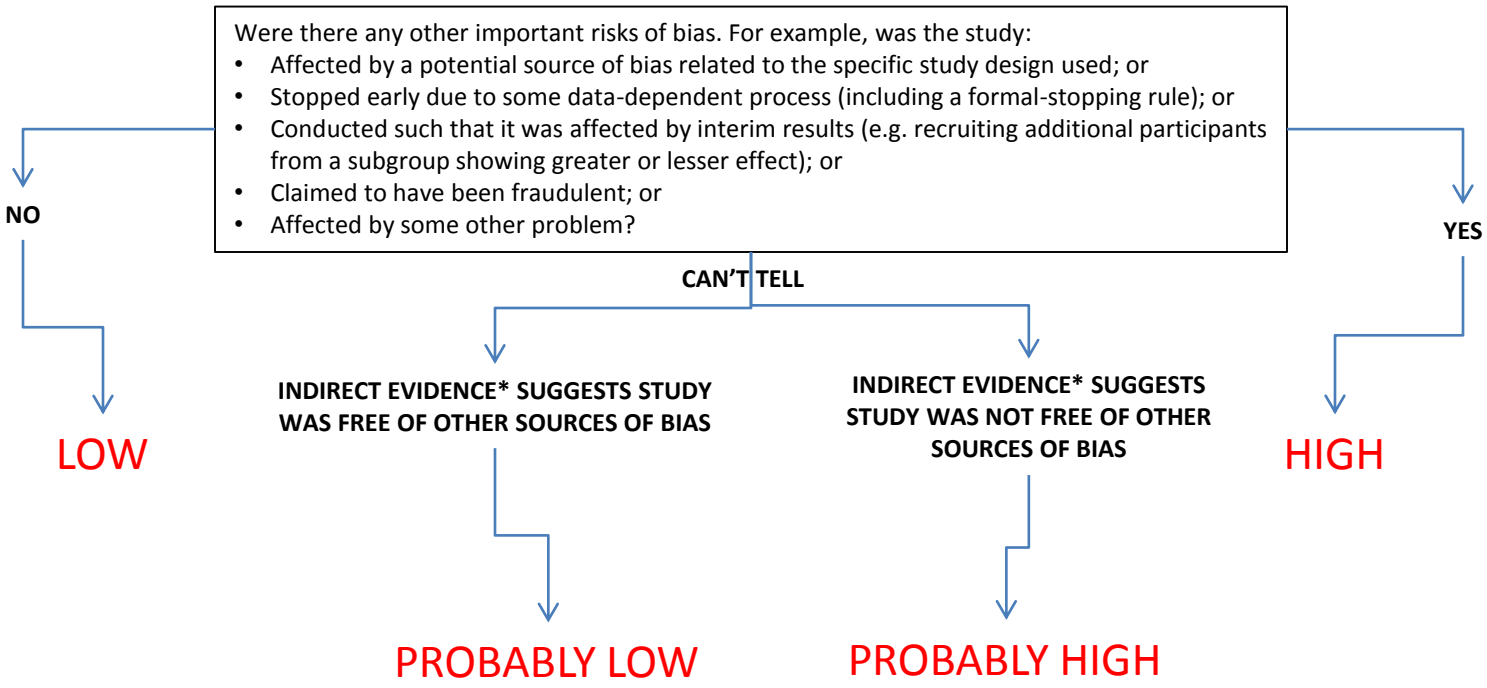
*e.g. based on study design or nature of outcomes

Conflict of Interest



*reviewer judgement

Other Sources of Bias



*reviewer judgement